ERIC THRANE

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EDUCATION

University of Washington, Seattle, WA *2003–2008*
PhD, Physics
A Search for Astrophysical Neutrino Point Sources with Super-Kamiokande
Advisor: R Jeffrey Wilkes

University of Michigan, Ann Arbor, MI *1999–2003*
BS, Physics with Highest Honors (& BA, Philosophy)
Flat Electron Beam Dynamics: A Comparison of Data with Simulation
Advisor: David Gerdes

RESEARCH INTERESTS

Astrophysics, gravitational waves, cosmology

WORK EXPERIENCE

Professor 2020–
School of Physics & Astronomy, Monash University *Clayton, VIC*

Chief Investigator 2024–2031
ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav 2) *Clayton, VIC*

Associate Professor 2018–2019
School of Physics & Astronomy, Monash University *Clayton, VIC*

Data Theme Leader 2017–2024
ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav) *Clayton, VIC*

Senior Lecturer 2017–2018
School of Physics & Astronomy, Monash University *Clayton, VIC*

Lecturer 2015–2016
School of Physics & Astronomy, Monash University *Clayton, VIC*

Senior Postdoctoral Scholar 2012–2014
Division of Physics, California Institute of Technology *Pasadena, CA*

Postdoctoral Research Associate 2008–2012
Dept. of Physics & Astronomy, University of Minnesota *Minneapolis, MN*

AWARDS & FELLOWSHIPS

ARC Linkage Infrastructure, Equipment and Facilities (LIEF; LE260100008) 2025

· Breakthrough Science in the Advanced LIGO+ Era ($3M AUD for 13 investigators)

Probing CP violation with Hyper-K 2025–2028

· $634K AUD with Prof Phil Urquijo; DP250100373

A Transdimensional Approach to Gravitational-Wave Astronomy 2023–2026

· $460K AUD with AProf Paul Lasky; DP230103088

The ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav 2) 2024–2031

· $35M AUD for 23 researchers; CE230100016

ARC Linkage Infrastructure, Equipment and Facilities (LIEF; LE210100002) 2021

· Australian Partnership in Advanced LIGO+ ($3M AUD for 12 investigators)

Rising stars (*The Australian*) 2019

· Australia's top 40 researchers who are less than 10 years into their careers

The ARC Centre of Excellence for Gravitational Wave Discovery (OzGrav) 2016–2023

· $31M AUD for 19 researchers; CE170100004

Breakthrough Prize in Fundamental Physics 2016

· $2M USD split between members of the LIGO Scientific Collaboration

Gruber Cosmology Prize 2016

· Ron Drever, Kip Thorne, Rai Weiss, and the LIGO Science Collaboration

ARC Future Fellowship (FT150100281) 2015–2019

· Gravitational-wave astronomy: detection and beyond ($618K AUD)

Ken Young Fellow 2003
University of Washington *Seattle, WA*

Graduated with Highest Honors 2003
University of Michigan *Ann Arbor, MI*

SELECT PUBLICATIONS

With significant personal contribution; my group members are highlighted in bold.

★ = lead and/or corresponding author

[1] M. L. Cross-Parkin, I. Magaña Hernandez, A. Palmese, P. D. Lasky, and E. Thrane, *Hostbusters: constraining H_0 and binary formation channels through gravitational-wave host galaxy associations*, (2026) arxiv/2610.25866.

[2] K. Grunthal, E. Thrane, R. van Haasteren, D. J. Champion, and M. Kramer, *Not just a phase: detecting nanohertz gravitational waves from phase alone*, 2026. arxiv/2610.06178.

[3] **L. Pinchbeck**, E. Thrane, C. Balazs, and P. Lasky, *`BilbyFlow`: user-friendly neural posterior estimation for gravitational-wave astronomy*, *Accepted in Astrophys. J. Supp.* (2026) arxiv/2609.00766.

[4] **B. Houlden**, E. Thrane, and K. Auchettl, *The properties of the first continuous gravitational waves: Optimizing pulsar timing observations for supermassive black hole binary detection*, (2026) arxiv/2608.25715.

[5] **N. Guttman**, P. D. Lasky, and E. Thrane, *Revealing four subpopulations of binary black-hole mergers with the fifth gravitational-wave transient catalog*, (2026) arxiv/2607.22011.

[6] **S. Y. Cheung**, R. Udall, D. Davis, P. D. Lasky, and E. Thrane, *Joint inference for gravitational-wave signal and noise glitch: Method and application*, *Accepted in Phys. Rev. D* (2026) arxiv/2607.09111.

[7] **C. Adamcewicz**, H. McDougall, P. D. Lasky, and E. Thrane, *Fortifying gravitational-wave population inference with normalizing flows*, *Phys. Rev. D* **114** (2026) 043057.

[8] **N. Guttman**, A. M. Baker, P. D. Lasky, and E. Thrane, *Licence to bin: Accurate and scalable inference for binary neutron stars in next-generation gravitational-wave detectors*, *Accepted in Phys. Rev. D* (2026) arxiv/2606.14197.

[9] Abac *et al.*, *GWTC-5.0: Population Properties of Merging Compact Binaries*, (2025) arxiv/2605.27226.

[10] **T. A. Clarke**, I. M. Romero-Shaw, C. Hoy, J. Stegmann, P. D. Lasky, and E. Thrane, *A universal framework to identify eccentric binary mergers: GW200105 case study*, (2026) arxiv/2605.18742.

[11] **M. Curyło**, E. Thrane, P. D. Lasky, and D. S. Gaynor, *A comprehensive framework for phase-coherent mapping of the gravitational-wave sky with pulsar timing arrays*, *Astrophys. J.* **1009** (2026) 166.

[12] **A. Atta**, N. Prouse, S. Chen, K. Okumura, P. de Perio, E. Thrane, and P. Urquijo, *Enhancing Event Reconstruction in Hyper-Kamiokande with Machine Learning: A ResNet Implementation*, (2026) arxiv/2604.13503.

[13] **V. Di Marco**, **N. Guttman**, M. T. Miles, A. Zic, R. M. Shannon, and E. Thrane, *A transdimensional sampling framework for pulsar timing noise modelling*, *Astrophys. J.* **1008** (2026) 234.

[14] **M. Sinha**, T. Clarke, M. Zhang, N. Sarin, E. Thrane, and P. Lasky, *Searching for gravitational waves from compact binary mergers powering long gamma-ray bursts during LIGO-Virgo-KAGRA's O3 run*, *Accepted in Mon. Not. R. Ast. Soc.* (2026) arxiv/2602.05169.

[15] K. Grunthal, D. J. Champion, E. Thrane, **R. S. Nathan**, M. Kramer, and M. T. Miles, *Optimising gravitational-wave sky maps for pulsar timing arrays*, *Astron. Astrophys.* **706** (2026) A352.

[16] **G. Curyło** *et al.*, *Frequency- and phase-resolved polarimetry of millisecond pulsars and its application to timing*, *Mon. Not. R. Ast. Soc.* **549** (2026) 1.

[17] **H. Tong**, T. A. Callister, M. Fishbach, E. Thrane, F. Antonini, S. Stevenson, I. M. Romero-Shaw, and F. Dosopoulou, *A subpopulation of low-mass, spinning black holes: signatures of dynamical assembly*, *Accepted in Phys. Rev. D* (2025) arxiv/2511.05316.

[18] A. G. Abac *et al.*, *GW241011 and GW241110: Exploring Binary Formation and Fundamental Physics with Asymmetric, High-spin Black Hole Coalescences*, *Phys. Rev. Lett.* **993** (2025) L21.

[19] A. G. Abac *et al.*, *Directional Search for Persistent Gravitational Waves: Results from the First Part of LIGO-Virgo-KAGRA's Fourth Observing Run*, (2025) arxiv/2510.17487.

[20] **L. Passenger**, **S. Banagiri**, E. Thrane, P. D. Lasky, *et al.*, *Is GW231123 a hierarchical merger?*, *Astrophys. J.* **999** (2026) 236.

[21] A. Ray, **S. Banagiri**, E. Thrane, and P. D. Lasky, *GW231123: extreme spins or microglitches?* , (2025) .

[22] A. G. Abac *et al.*, *GW250114: Testing Hawkings Area Law and the Kerr Nature of Black Holes*, *Phys. Rev. Lett.* **135** (2025) 111403.

[23] **H. Tong**, M. Fishbach, E. Thrane, *et al.*, *Evidence of the pair instability gap in the distribution of black hole masses*, *Nature* **652** (2026) 1.

[24] **N. Guttman**, E. Payne, P. D. Lasky, and E. Thrane, *Trends in the Population of Binary Black Holes Following the Fourth Gravitational-Wave Transient Catalog: a Data-Driven Analysis*, *Astrophys. J.* **996** (2026) 144.

[25] **S. Banagiri**, E. Thrane, and P. D. Lasky, *Evidence for three subpopulations of merging binary black holes at different primary masses*, *Phys. Rev. Lett.* **137** (2026) 021403 Editor's Suggestion.

[26] **C. Adamcewicz**, **N. Guttman**, P. D. Lasky, and E. Thrane, *Do both black holes spin in merging binaries? evidence from gwtc-4 and astrophysical implications*, *Astrophys. J.* **994** (2025) 261.

[27] Abac *et al.*, *GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run*, (2025) arxiv/2508.18082.

[28] Abac *et al.*, *GWTC-4.0: Population Properties of Merging Compact Binaries*, (2025) arxiv/2508.18083.

[29] Abac *et al.*, *GW231123: a Binary Black Hole Merger with Total Mass 190-265 $M_\odot$*, *Astrophys. J. Lett.* **993** (2025) .

[30] **L. Passenger**, **S.-Y. Cheung**, **N. Guttman**, **N. Kannachel**, P. D. Lasky, and E. Thrane, *A gaussian process framework for testing general relativity with gravitational waves*, *Astrophys. J.* **994** (2025) 202.

[31] **S.-Y. Cheung**, **L. Passenger**, P. D. Lasky, and E. Thrane, *A search for extra polarisations using a gaussian process in gravitational-wave transient catalogue 3*, *Astrophys. J.* **1001** (2026) 130.

[32] X.-X. Kou, M. Saleem, V. Mandic, C. Talbot, and E. Thrane, *Progress toward the detection of the gravitational-wave background from stellar-mass binary black holes: a mock data challenge*, *Phys. Rev. D* **112** (2025) 084064.

[33] **A. M. Baker**, P. D. Lasky, E. Thrane, and J. Golomb, *Significant challenges for astrophysical inference with next-generation gravitational-wave observatories*, *Phys. Rev. D* **112** (2025) 102004.

[34] **H. Tong**, M. Fishbach, and E. Thrane, *Spinning spectral sirens: Robust cosmological measurement using mass-spin correlations in the binary black hole population*, *Astrophys. J.* **98** (2025) 220.

[35] **V. Di Marco**, A. Zic, R. M. Shannon, E. Thrane, and A. D. Kulkarni, *Choosing suitable noise models for nanohertz gravitational-wave astrophysics*, *Astrophys. J.* **990** (2025) 85.

[36] **T. A. Clarke**, P. D. Lasky, and E. Thrane, *Inferring jet physics from neutron star - black hole mergers with gravitational waves*, *Astrophys. J.* **984** (2025) 27.

[37] **N. Guttman**, P. D. Lasky, and E. Thrane, *Modelling noise in gravitational-wave observatories with transdimensional models*, *Phys. Rev. D* **111** (2025) 063063.

[38] **L. Pinchbeck**, C. Balazsa, and E. Thrane, *Model-independent dark matter detection with the Cherenkov Telescope Array Observatory*, *J. Cosmo. R. Ast. Part.* **2026** (2026) 029.

[39] K. Grunthal, **R. S. Nathan**, *et al.*, *The MeerKAT pulsar timing array: Maps of the gravitational-wave sky with the 4.5 year data release*, *Mon. Not. R. Ast. Soc.* **536** (2024) 1501.

[40] M. Miles *et al.*, *The MeerKAT Pulsar Timing Array: The first search for gravitational waves with the MeerKAT radio telescope*, *Mon. Not. R. Ast. Soc.* (2024) .

[41] Z.-Q. You *et al.*, *Determination of the birth-mass function of neutron stars from observations*, *Nat. Astro.* **457** (2025) 2397.

[42] **S. R. Goode**, M. Schiworski, D. Brown, E. Thrane, and P. D. Lasky, *You only thermoelastically deform once: Point Absorber Detection in LIGO Test Masses with YOLO*, *Optics Express* **33** (2025) 17601 Featured in Spotlight on Optics.

[43] **T. A. Clarke**, N. Sarin, E. J. Howell, P. D. Lasky, and E. Thrane, *Quantifying the coincidence between gravitational waves and fast radio bursts from neutron star–black hole mergers*, *Phys. Rev. D* **11** (2025) 083023.

[44] **C. Adamcewicz**, P. D. Lasky, E. Thrane, and I. Mandel, *No evidence for a dip in the binary black hole mass spectrum*, *Astrophys. J.* **975** (2024) 253.

[45] **L. Passenger**, E. Thrane, P. D. Lasky, E. Payne, S. Stevenson, and B. Farr, *Are all models wrong? falsifying binary formation models in gravitational-wave astronomy*, *Mon. Not. R. Ast. Soc.* **535** (2024) 2837.

[46] **S. Y. Cheung**, P. D. Lasky, and E. Thrane, *Does spacetime have memories? Searching for gravitational-wave memory in the third LIGO-Virgo-KAGRA gravitational-wave transient catalogue*, *Class. Quantum Grav.* **41** (2024) 115010.

[47] The LVK Collaborations, *Observation of gravitational waves from the coalescence of a* $2.5 - 4.5 M_\odot$ *compact object and a neutron star*, *Astrophys. J. Lett.* **970** (2024) L34.

[48] **H. Tong** *et al.*, *Transdimensional inference for gravitational-wave astronomy with* `Bilby`, *Astrophys. J. Supp.* **276** (2025) 50.

[49] **V. Di Marco**, A. Zic, R. M. Shannon, and E. Thrane, *Systematic errors in searches for nanohertz gravitational waves*, *Mon. Not. R. Ast. Soc.* **532** (2024) 4026.

[50] **T. A. Clarke**, M. Isi, P. D. Lasky, E. Thrane, *et al.*, *Striking the right tone: towards a self-consistent framework for measuring black hole ringdowns*, *Phys. Rev. D* **109** (2024) 124030.

[51] **L. Pinchbeck**, E. Thrane, and C. Balazs, *GammaBayes: a Bayesian pipeline for dark matter detection with CTA*, *J. Cosmo. R. Ast. Part.* **2024** (2024) 020.

[52] **K. Walker**, **R. Smith**, E. Thrane, and D. J. Reardon, *Precision constraints on the neutron star equation of state with third-generation gravitational-wave observatories*, *Phys. Rev. D* **110** (2024) 043013.

[53] **C. Adamcewicz**, P. D. Lasky, and E. Thrane, *Which black hole is spinning? probing the origin of black-hole spin with gravitational waves*, *Astrophys. J. Lett.* **964** (2024) L6.

[54] J. W. Gardner, L. Sun, S. Borhanian, P. D. Lasky, E. Thrane, D. E. McClelland, and B. J. J. Slagmolen, *Multi-messenger astronomy with a southern-hemisphere gravitational-wave observatory*, *Phys. Rev. D* **108** (2023) 123026.

[55] **C. Adamcewicz**, P. D. Lasky, and E. Thrane, *Evidence for a correlation between binary black hole mass ratio and black-hole spins*, *Astrophys. J.* **958** (2023) 13.

[56] D. J. Reardon *et al.*, *Search for an isotropic gravitational-wave background with the parkes pulsar timing array*, *Astrophys. J. Lett.* **951** (2023) L6.

[57] **V. Di Marco**, A. Zic, M. T. Miles, D. J. Reardon, E. Thrane, and R. M. Shannon, *Toward robust detections of nanohertz gravitational waves*, *Astrophys. J.* **956** (2023) 14.

[58] **R. S. Nathan** *et al.*, *Improving pulsar-timing solutions through dynamic pulse fitting*, *Mon. Not. R. Ast. Soc.* **523** (2023) 4405.

[59] B. Allen *et al.*, *The international pulsar timing array checklist for the detection of nanohertz gravitational waves*, (2023) arxiv/2304.04767 ★.

[60] **T. A. Clarke**, L. Chastain, P. D. Lasky, and E. Thrane, *Nuclear physics with gravitational waves from neutron stars disrupted by black holes*, *Astrophys. J. Lett.* **949** (2023) L6.

[61] E. Payne and E. Thrane, *Model exploration in gravitational-wave astronomy with the maximum population likelihood*, *Phys. Rev. Res.* **5** (2023) 023013.

[62] J. Paynter and E. Thrane, *Meet the parents: the progenitor binary for the supermassive black hole candidate in E1821+643*, *Astrophys. J. Lett.* **945** (2023) L18.

[63] **H. Tong**, **S. Galaudage**, and E. Thrane, *The population properties of spinning black holes using Gravitational-wave Transient Catalog 3*, *Phys. Rev. D* **106** (2022) 103019.

[64] **C. Adamcewicz** and E. Thrane, *Do unequal-mass binary black hole systems have larger χ_{eff}? Probing correlations with copulas in gravitational-wave astronomy*, *Mon. Not. R. Ast. Soc.* **517** (2022) .

[65] A. M. Knee, **I. M. Romero-Shaw**, P. D. Lasky, J. McIver, and E. Thrane, *A Rosetta Stone for eccentric gravitational waveform models*, *Astrophys. J.* **936** (2022) 172.

[66] **I. Romero-Shaw**, P. Lasky, and E. Thrane, *Four eccentric mergers increase the evidence that ligo–virgo–kagra's binary black holes form dynamically*, *Astrophys. J.* **940** (2022) 171.

[67] **T. A. Clarke**, **I. M. Romero-Shaw**, P. D. Lasky, and E. Thrane, *Gravitational-wave inference for eccentric binaries: the argument of periapsis*, *Mon. Not. R. Ast. Soc.* **517** (2022) 3778.

[68] S. Biscoveanu, K. Kremer, and E. Thrane, *Probing the efficiency of tidal synchronization in outspiralling double white dwarf binaries with lisa*, *Astrophys. J.* **949** (2023) 95.

[69] B. Goncharov *et al.*, *Consistency of the PPTA signal with a nanohertz gravitational-wave background*, *Astrophys. J. Lett.* **932** (2022) L22.

[70] F. Broekgaarden, S. Stevenson, and E. Thrane, *Signatures of mass ratio reversal in gravitational waves from merging binary black holes*, *Astrophys. J.* **938** (2022) 45.

[71] **A. Makai** Baker, P. D. Lasky, E. Thrane, *et al.*, *GWCloud: a searchable repository for the creation and curation of gravitational-wave inference results*, *Astrophys. J. Supp.* **266** (2023) 33.

[72] **K. Walker**, D. J. Reardon, E. Thrane, and **R. Smith**, *Orbital dynamics and extreme scattering event properties from long-term scintillation observations of psr j16037202*, *Astrophys. J.* **933** (2022) 16.

[73] **A. Vajpeyi**, **R. Smith**, and E. Thrane, *Deep follow-up of GW151226: ordinary binary or low-mass-ratio system?*, *Astrophys. J.* **947** (2023) 10.

[74] **I. M. Romero-Shaw**, E. Thrane, and P. D. Lasky, *When models fail: an introduction to posterior predictive checks and model misspecification in gravitational-wave astronomy*, *Pub. Astron. Soc. Aust.* **39** (2022) E025 arxiv/2202.05479.

[75] N. Sarin, P. D. Lasky, **F. H. Vivanco**, S. P. Stevenson, D. Chattopadhyay, **R. Smith**, and E. Thrane, *Linking the rates of neutron star binaries and short gamma-ray bursts*, *Phys. Rev. D* **105** (2022) 083004.

[76] **A. Mangipudi**, E. Thrane, and C. Balazs, *Bayesian WIMP detection with the Cherenkov Telescope Array*, *J. Cosmo. R. Ast. Part.* **2022** (2022) 010.

[77] V. Kalogera *et al.*, *The Next Generation Global Gravitational Wave Observatory: The Science Book*, (2021) arxiv/2111.06990.

[78] R. Abbott *et al.*, *GWTC-3: Compact Binary Coalescences Observed by LIGO and Virgo During the Second Part of the Third Observing Run*, *Phys. Rev. X* **13** (2023) 041039.

[79] R. Abbott *et al.*, *The population of merging compact binaries inferred using gravitational waves through GWTC-3*, *Phys. Rev. X* **13** (2023) 011048.

[80] R. Abbott *et al.*, *Constraints on the cosmic expansion history from GWTC-3*, *Astrophys. J.* **949** (2023) 76.

[81] **A. Vajpeyi**, E. Thrane, **R. Smith**, B. McKernan, and K. S. Ford, *Measuring the properties of active galactic nuclei disks with gravitational waves*, *Astrophys. J.* **931** (2022) 82.

[82] P. D. Lasky and E. Thrane, *Did goryachev et al. detect megahertz gravitational waves?*, *Phys. Rev. D* **104** (2021) 103017.

[83] **S. Galaudage**, C. Talbot, **T. Nagar**, D. Jain, E. Thrane, and I. Mandel, *Building better spin models for merging binary black holes: Evidence for non-spinning and rapidly spinning nearly aligned sub-populations*, *Astrophys. J. Lett.* **921** (2021) L15.

[84] R. Essick, A. Farah, **S. Galaudage**, C. Talbot, M. Fishbach, E. Thrane, and D. E. Holz, *Don't just leave-one-out: Probing extremal gravitational-wave events with coarse-grained likelihoods*, *Astrophys. J.* **926** (2022) 34.

[85] **I. M. Romero-Shaw**, P. D. Lasky, and E. Thrane, *Signs of eccentricity in two gravitational-wave signals may indicate a sub-population of dynamically assembled binary black holes*, *Astrophys. J. Lett.* **921** (2021) L31.

[86] **E. Payne**, L. Sun, K. Kremer, P. D. Lasky, and E. Thrane, *The imprint of superradiance on hierarchical black hole mergers*, *Astrophys. J.* **931** (2022) 79.

[87] B. Goncharov *et al.*, *On the evidence for a common-spectrum process in the search for the nanohertz gravitational wave background with the Parkes Pulsar Timing Array*, *Astrophys. J. Lett.* **917** (2021) L19.

[88] **A. Vajpeyi**, **R. Smith**, E. Thrane, *et al.*, *A search for intermediate-mass black holes mergers in the second LIGO–Virgo observing run with the Bayes Coherence Ratio*, *Mon. Not. R. Ast. Soc.* **516** (2022) 5309.

[89] B. McKernan, K. E. S. Ford, T. Callister, W. M. Farr, R. O'Shaughnessy, **R. Smith**, E. Thrane, and **A. Vajpeyi**, *LIGO–Virgo correlations between mass ratio and effective inspiral spin: testing the active galactic nuclei channel*, *Mon. Not. R. Ast. Soc.* **514** (2022) 3886.

[90] **R. Willcox**, I. Mandel, E. Thrane, A. Deller, S. Stevenson, and A. Vigna-Gómez, *Constraints on weak supernova kicks from observed pulsar velocities*, *Astrophys. J. Lett.* **920** (2021) L37.

[91] R. Abbott *et al.*, *Observation of gravitational waves from two neutron star–black hole coalescences*, *Astrophys. J. Lett.* **915** (2021) L5.

[92] C. Talbot, E. Thrane, S. Biscoveanu, and **R. Smith**, *Inference with finite time series: Observing the gravitational Universe through windows*, *Phys. Rev. Res.* **3** (2021) 043049.

[93] M. Zevin, **I. M. Romero-Shaw**, K. Kremer, E. Thrane, and P. D. Lasky, *Implications of eccentric observations on binary black hole formation channels*, *Astrophys. J. Lett.* **921** (2021) L43.

[94] R. Abbott *et al.*, *Constraints on Cosmic Strings Using Data from the Third Advanced LIGO–Virgo Observing Run*, *Phys. Rev. Lett.* **126** (2021) 241102.

[95] **Z.-Q. You**, G. Ashton, **X.-J. Zhu**, E. Thrane, and Z.-H. Zhu, *Optimized localization for gravitational-waves from merging binaries*, *Mon. Not. R. Ast. Soc.* **509** (2021) 3957.

[96] **M. Hübner**, P. D. Lasky, and E. Thrane, *Memory remains undetected: Updates from the second LIGO/Virgo gravitational-wave transient catalog*, *Phys. Rev. D* **104** (2021) 023004.

[97] J. Paynter, R. Webster, and E. Thrane, *Evidence for an intermediate-mass black hole from a gravitationally lensed gamma-ray burst*, *Nat. Astron.* **5** (2021) 560.

[98] **R. Smith** *et al.*, *Bayesian inference for gravitational waves from binary neutron star mergers in third-generation observatories*, *Phys. Rev. Lett.* **127** (2021) 081102.

[99] **C. Talbot** and E. Thrane, *Fast, flexible, and accurate evaluation of malmquist bias with machine learning: Preparing for the pending flood of gravitational-wave detections*, *Astrophys. J.* **927** (2022) 76.

[100] **I. M. Romero-Shaw**, K. Kremer, P. D. Lasky, E. Thrane, and J. Samsing, *Gravitational waves as a probe of globular cluster formation and evolution*, *Mon. Not. R. Ast. Soc.* **506** (2021) 2362.

[101] C. Kimball, C. Talbot, C. P. Berry, M. Zevin, E. Thrane, *et al.*, *Evidence for hierarchical black hole mergers in the second LIGO–Virgo gravitational-wave catalog*, *Astrophys. J. Lett.* **915** (2021) L35.

[102] **S. Galaudage**, **C. Adamcewicz**, **X.-J. Zhu**, S. Stevenson, and E. Thrane, *Heavy double neutron stars: birth, mid-life and death*, *Astrophys. J. Lett.* **909** (2021) L19.

[103] C. D. Blair, Y. Levin, and E. Thrane, *Constraining temperature distribution inside LIGO test masses from frequencies of their vibrational modes*, *Phys. Rev. D* **103** (2021) 022003.

[104] R. Abbott *et al.*, *GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run*, *Phys. Rev. X* **11** (2021) 021053.

[105] R. Abbott *et al.*, *Population properties of compact objects from the second LIGO-Virgo Gravitational-Wave Transient Catalog*, *Astrophys. J. Lett.* **913** (2021) L7 Focus Issue: Gravitational-wave Astrophysics from the Second LIGO-Virgo Transient Catalog.

[106] **B. Goncharov** *et al.*, *Identifying and mitigating noise sources in precision pulsar timing data sets*, *Mon. Not. R. Ast. Soc.* **502** (2020) 478.

[107] J. Calderón Bustillo, P. D. Lasky, and E. Thrane, *Black-hole spectroscopy, the no-hair theorem and GW150914: Kerr vs. Occam*, *Phys. Rev. D* **103** (2021) 024041.

[108] **E. Payne**, C. Talbot, P. D. Lasky, E. Thrane, and J. S. Kissel, *Gravitational-wave astronomy with a physical calibration model*, *Phys. Rev. D* **102** (2020) 122004.

[109] **I. M. Romero-Shaw**, P. D. Lasky, E. Thrane, and J. Calderón Bustillo, *GW190521: orbital eccentricity and signatures of dynamical formation in a binary black hole merger signal*, *Astrophys. J. Lett.* **903** (2020) L5 Norris Family Publication Award.

[110] **S. Biscoveanu**, **C. Talbot**, E. Thrane, and **R. Smith**, *Measuring the primordial gravitational-wave background in the presence of astrophysical foregrounds*, *Phys. Rev. Lett.* **125** (2020) 241101.

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TELESCOPE PROPOSALS

MeerKAT (PI: R. Shannon) — 588 hours over two years
The MeerKAT Pulsar Timing Array — *2025–2027*

RECENT TALKS

Australian National University RSAA — June 2024
Results from the 4.5 year MeerKAT Pulsar Timing Array Data Release — *Mt Stromlo, ACT*

IPTA Meeting — June 2024
Mapping the nanohertz gravitational-wave sky — *Sexton, Italy*

GWADW — May 2024
Astrophysics in the Era of neXt-Generation Observatories — *Hamilton Island, QLD*

IAU-IAA Astrostats Seminar — August 2022
Investigating black hole spin with gravitational waves — *Online*

Frontiers of Fundamental Physics — May 2022
Merging compact binaries inferred using gravitational waves through GWTC-3 — *Istanbul, Turkey*

Association of Asia Pacific Physical Societies — October 2021
Building better spin models for merging binary black holes — *Seoul, South Korea*

University of Melbourne — September 2021
Building better models for populations of merging binary black holes — *Melbourne, VIC*

University of Michigan — March 2021
Population Properties of Compact Objects from GWTC-2 — *Ann Arbor, MI*

University of New South Wales — December 2020
Compact objects in the Second LIGO-Virgo Gravitational-wave Transient Catalog — *Sydney, NSW*

University of Canterbury — December 2020
Population Properties from the Second LIGO-Virgo Catalog — *Canterbury, NZ*

LIGO-Virgo Webinar — November 2020
Population Properties of Compact Objects from the Second LIGO-Virgo Catalog — *Online*

University of Auckland — October 2020
The population properties of binary black holes with Bayesian hierarchical modelling — *Auckland, NZ*

CSIRO Astronomy & Space Science — July 2020
Dispatches from the black hole mass gaps: recent results from LIGO-Virgo — *Marsfield, NSW*

LEADERSHIP & SERVICE

Referee

Astronomy & Astrophysics, Astrophysical Journal, Astrophysical Journal Letters, Journal for Cosmology and Astroparticle Physics, Living Reviews in Relativity, Monthly Notices of the Royal Astronomical Society, New Astronomy Reviews, Nature Astronomy, Nature Communications, Physical Review Applied, Physical Review D, Physical Review Letters

Reviewer

- Australian Research Council (ARC), Dutch Research Council, Royal Society Te Apārangi (New Zealand), Swiss National Science Foundation, US National Science Foundation (NSF)

Advisory

- National Committee for Astronomy (NCA) Member (2026–), NCA Time Domain and Multi-Messenger Astrophysics Working Group Chair (2024), AAL Project Oversight Committee (2024–2025), ASA Time-Domain Astronomy Steering Committee, IPTA Detection Committee (2021–2023), NCA MTR CapOp (2019), AAL Science Advisory (2018–2020)

LIGO Scientific Collaboration

- Co-Chair of Stochastic Data-Analysis Group (2011–2017), Review Chair for Burst Group (2017–2020), Editorial Board (2019–), Co-Chair of Compact Binary Coalescence Group (2025)

Organising Committees

- ASA SOC 2019; GWPAW SOC (2018, 2021, 2022); AGCGRG SOC 2021; GWPAW LOC 2022

Diversity

- LVC Ally (2018–), OzGrav Diversity Committee (2017–)

MEDIA

My group's work has been featured in a number of publications including *The Independent*, *The Sydney Morning Herald*, *The Australian*, CNET, and *The Guardian*. I have discussed my research on radio and television programs including *The 7:30 Report*, *Catalyst*, 3AW radio.

OUTREACH

I regularly give public lectures on black holes and gravitational waves. Highlights include MIT Lincoln Lab, the AIP Nobel Prize Public Lecture, and an Instant Expert event organised by New Scientist.

TEACHING & EDUCATION

Monash University 2015–

- PHS1011 (First-Year Physics): Unit Coordinator
- PHS1022 (First-Year Physics): Unit Coordinator
- ASP2062 (Astrophysics): Unit Coordinator
- PHS4200 (General Relativity)
- PHS5020 (Advanced General Relativity)

Administrative roles

- Postgraduate Research Coordinator 2022–2024
- Education Head 2015–2016

Supervision

· Research faculty
  - Dr Rory Smith 2017–2024

· Postdocs
  - Dr Gosia Curyło 2024–
  - Dr Sharan Banagiri 2024–
  - Dr Nir Guttman 2023–
  - Dr Simon Goode 2023–
  - Dr Grant Meadors 2018–2019
  - Dr Xingjiang Zhu 2017–2021
  - Dr Letizia Sammut 2015–2017
  - Dr Pablo Rosado 2016

· Postgraduate Students
  - Blaze Houlden 2025–
  - Mallika Sinha 2025–
  - Andrew Atta 2024–
  - Lachlan Passenger 2024–
  - Liam Pinchbeck 2023–
  - Shun Cheung 2023–
  - Hui Tong 2022–
  - Christian Adamcewicz 2022–
  - Teagan Clarke 2022–2025
  - Valentina DiMarco 2022–2025
  - Rowina Nathan 2022–2025
  - Avi Vajpeyi 2020–2023
  - Shanika Galaudage 2019–2023
  - Isobel Romero-Shaw 2018–2021
    - Winner: Robert Street Doctoral Prize
  - Moritz Hübner 2018–2021
  - Francisco Hernandez 2017–2021
  - Colm Talbot 2016–2020
    - Winner: Vice-Chancellors Commendation for Thesis Excellence
    - Winner: Charlene Heisler Prize for the most outstanding astronomy PhD thesis in Australia
    - Winner: Robert Street Doctoral Prize
  - Boris Goncharov 2016–2020
  - Sylvia Biscoveanu (Fulbright) 2017–2018

· Honours/MSc Students
  - Makai Baker 2024
  - Chan Anand 2024
  - Michelle Zhang 2024
  - Lachlan Passenger 2023
  - Nikhil Kannachel 2023
  - Jiaxuan Zhou 2022
  - Macauley Angus 2022
  - Tushar Nagar 2021
  - Kris Walker 2021

- Abhi Mangipudi 2020–2021
- Ethan Payne 2020
  - Winner: Australian Institute of Physics Laby Medal for the best Honours or Masters thesis from an Australian University
- Nick Farrow 2019
- Marcus Lower 2017
- Chris Whittle 2016
- Lucy McNeill 2016

· Undergraduate Students

- Emma Sapkin 2024
- Ella Garth 2024
- Jordan Klein 2022
- Carter Hills 2020
- Atul Divakarla (IREU from Florida) 2018
- Alex Kemp 2017
- William Campbell 2015
- Shi Qiu 2015
- Tyson Jones 2015